

Expt. No. _____

Statistics - Module 5.

■ Correlation: The statistical technique which ~~may~~ can be used to study the relationship between 2 or more variables is called correlation analysis.

eg: ① when the price of a commodity increases, the supply ^{for} that commodity decreases

② Different types of correlation

1] positive and negative correlation:-

when the variables are varying in the same direction it is called +ve correlation.

(if $x \uparrow$ then $y \uparrow$ or if $x \downarrow$ then $y \downarrow$)

when the variables are varying in the opposite direction it is called -ve correlation.

(if $x \uparrow, y \downarrow$ or if $x \downarrow, y \uparrow$)

Eg:-

x:	10	20	30	40	50
y:	80	100	150	170	200

} +ve correlation.

x:	40	30	25	20	10
y:	2	3	5	8	12

} -ve correlation.

2] Linear and Non linear correlation

if the ratio of change between the two sets of variables is same then it is called linear correlation.

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eg: X: 10 15 30 60
Y: 50 75 150 300

If the amount of change in one variable does not bring the same ratio of change in the other variable it is called Nonlinear.

eg: X: 2 4 6 10 15
Y: 8 10 18 22 26

3] Perfect and imperfect correlation

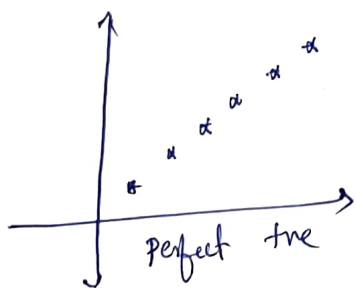
The correlation b/w 2 variables is usually denoted by r

The value of r lies between -1 to 1

If the value of correlation between the 2 variables x and y is 1 or -1 , we say that there is perfect correlation. If the value of r is in between -1 & 1 we say that imperfect correlation.

Measures of correlation

1) Scatter Diagram: The simplest method for ascertaining the correlation between variables.



2) Karl Pearson's Coefficient of correlation

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

Q. calculate Karl Pearson's coefficient of correlation for the following data

x: 2 3 4 5 6 7 8
y: 4 5 6 12 9 5 4

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

x	y	xy	x ²	y ²
2	4	8	4	16
3	5	15	9	25
4	6	24	16	36
5	12	60	25	144
6	9	54	36	81
7	5	35	49	25
8	4	32	64	16
<u>35</u>	<u>45</u>	<u>228</u>	<u>203</u>	<u>343</u>

$$= \frac{7 \times 228 - 35 \times 45}{\sqrt{7 \times 203 - (35)^2} \sqrt{7 \times 343 - (45)^2}}$$

$$= \frac{1596 - 1575}{14 \times 19.39}$$

$$= \underline{\underline{0.011}}$$

the correlation.

Q. calculate the coefficient of correlation for the following data

x: 51 54 55 59 65 60 70
y: 38 44 33 36 33 23 10

x	y	xy	x^2	y^2
51	38	1938	2601	1444
54	44	2376	2916	1936
55	33	1815	3025	1089
59	36	2124	3481	1296
65	33	2145	4225	1089
60	23	1380	3600	529
70	10	700	4900	100
414	217	12478	24748	7483

$$r = \frac{24748 - 414 \times 217}{\sqrt{7 \times (24748)^2 - 414^2} \times \sqrt{7 \times 7483 - 217^2}}$$

$$= \frac{83398}{\sqrt{1840} \times \sqrt{5292}}$$

$$= \frac{-2492}{3120.46}$$

$$= \underline{\underline{-0.798}}$$

3] Spearman's Rank correlation

case 1: when ranks are distinct

$$R = 1 - \frac{6 \sum D^2}{n^3 - n}$$

where D = Difference between
Ranks.

Ranks in maths : 1 2 3 4 Find Rank correlation
Ranks in Eng : 3 4 2 1

$R(x)$	$R(y)$	D	D^2
1	3	-2	4
2	4	-2	4
3	2	1	1
4	1	3	9
			18

$$\therefore R = 1 - \frac{6 \times 18}{4^3 - 4}$$

$$= 1 - \frac{6 \times 18}{60}$$

$$= 1 - \frac{108}{60} = 1 - 1.8$$

$$= \underline{\underline{-0.8}}$$

Q. calculate Rank coefficient for the following data of weight of father and son

weight of father : 54 54 55 59 65 60 70
 weight of son : 38 44 33 36 34 23 10

(Ans)

x	y	R(x)	R(y)	D	D ²
54	38	5	2	3	9
54	44	7	10	6	36
55	33	6	5	1	1
59	36	4	3	1	1
65	34	2	4	-2	4
60	23	3	6	-3	9
70	10	1	7	-6	36
					96

$$\therefore R = 1 - \frac{6 \sum D^2}{n^3 - n} = 1 - \frac{6 \times 96}{7^3 - 7} = 1 - \frac{576}{336} = 1 - 1.71 = \underline{\underline{-0.71}}$$

Case 2: when Ranks are repeated

$$R = 1 - \frac{6 \left[\sum D^2 + \sum \left(\frac{m^3 - m}{12} \right) \right]}{n^3 - n}$$

Q. calculate the rank correlation coefficient.

Mark in English : 80 45 55 58 55 60 45 68 70 45 85

Mark in Statistics : 82 56 50 43 56 62 64 65 70 64 90

x	y	$R(x)$	$R(y)$	D	D^2
80	82	2	2	0	0
45	56	10	8.5	1.5	2.25
55	50	7.5	10	-2.5	6.25
58	43	6	11	-5	25
55	56	7.5	8.5	-1	1
60	62	5	7	-2	4
45	64	10	5.5	4.5	20.25
68	65	4	4	0	0
70	70	3	3	0	0
45	64	10	5.5	4.5	20.25
85	90	1	1	0	0
					79

55 Repeats twice, $m^3 - m = 2^3 - 2 = 6$

45 Repeats three, $m^3 - m = 3^3 - 3 = 24$

66 Repeats twice, $m^3 - m = 2^3 - 2 = 6$

64 Repeats twice, $m^3 - m = 2^3 - 2 = 6$

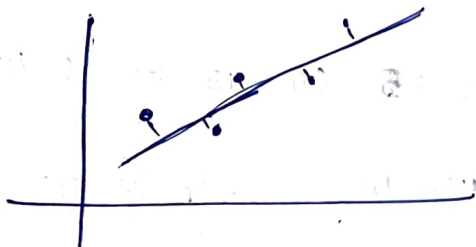
$$\therefore \frac{\sum m^3 - m}{12} = \frac{6 + 24 + 6 + 6}{12} = \frac{42}{12}$$

$$\therefore R = 1 - \frac{6 \left[79 + \frac{42}{12} \right]}{11^3 - 11} = 1 - \frac{6 \left[79 + 3.5 \right]}{1320}$$

$$= 1 - \frac{6 \left[82.5 \right]}{1320} = 1 - \frac{495}{1320} = \underline{\underline{0.625}}$$

The Principle of Least Square

Let $y = f(x)$ be a bivariate data, then principle of least square states that the sum of the squares of distance between the given points and points in the line should be the least



Fit a straight line $y = a + bx$

Normal equations are $\Sigma y = na + b \Sigma x$

$$\Sigma xy = a \Sigma x + b \Sigma x^2$$

using the principle of least square fit a straight line

$x = 5 \quad 10 \quad 15 \quad 20 \quad 25$

$y = 15 \quad 19 \quad 23 \quad 26 \quad 30$

x	y	x^2	y^2	xy
5	15	25	225	75
10	19	100	361	190
15	23	225	529	345
20	26	400	676	520
25	30	625	900	750
75	113	1375	2691	1880

$$113 = 5a + b \cdot 75$$

$$1880 = 75a + b \cdot 1375$$

i.e.,

$$\Rightarrow 5a + 75b = 113 \quad \text{--- (1)}$$

$$75a + 1375b = 1880 \quad \text{--- (2)}$$

$$\textcircled{1} \times 15 \Rightarrow 75a + 1125b = 1695 \quad \text{--- (3)}$$

$$75a + 1375b = 1880 \quad \text{--- (2)}$$

$$\textcircled{3} - \textcircled{2} \Rightarrow -250b = -185$$

$$b = 0.74$$

$$\therefore 5a + 75 \times 0.74 = 113 \Rightarrow 5a = 57.5 \Rightarrow a = 11.5$$

$$\therefore y = 11.5 + 0.74x$$

Fit a straight line $x = 2 \quad 3 \quad 4 \quad 5$
 $y = 7 \quad 9 \quad 10 \quad 11$

x	y	x^2	y^2	xy
2	7	4	49	14
3	9	9	81	27
4	10	16	100	40
5	11	25	121	55
14	37	54	351	136

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Fitting a parabola

$$y = a + bx + cx^2$$

Normal Equations are

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

Fit a parabola

$x: 1 \ 2 \ 3 \ 4 \ 5$

$y: 3 \ 9 \ 16 \ 25 \ 40$

x	y	x^2	x^3	x^4	xy	x^2y
1	3	1	1	1	3	3
2	9	4	8	16	18	36
3	16	9	27	81	48	144
4	25	16 16	64 64	256 256	100 100	400
5	40	25 25	125 125	625 625	200 200	1000
15	93	55	225	979	369	1583

$$\Rightarrow 5a + 15b + 55c = 93 \quad \text{--- (1)}$$

$$15a + 55b + 225c = 369 \quad \text{--- (2)}$$

$$55a + 225b + 979c = 1583 \quad \text{--- (3)}$$

$$\textcircled{1} \times 3 \Rightarrow 15a + 45b + 165c = 279$$

$$15a + 55b + 225c = 369$$

$$-10b - 60c = -90$$

$$\Rightarrow 10b + 60c = 90 \quad \text{--- (4)}$$

$$(2) \times 55 \Rightarrow 825a + 3025b + 1237c = 20295$$

$$(3) \times 15 \Rightarrow 825a + 3375b + 14685c = 23745$$

$$-350b - 2310c = -3450$$

$$\Rightarrow 350b + 2310c = 3450 \quad \text{--- (5)}$$

$$(4) \times 35 \Rightarrow 350b + 2100c = 3150$$

$$210c = 300$$

$$\Rightarrow \boxed{c = 1.428}$$

$$\text{Sub in (4)} \Rightarrow 10b + 60 \times 1.428 = 90$$

$$\Rightarrow 10b = 4.32 \Rightarrow \boxed{b = 0.432}$$

$$\text{Sub in (1)} \Rightarrow 5a + 15 \times 0.432 + 55 \times 1.428 = 93$$

$$\Rightarrow 5a + 6.48 + 78.54 = 93$$

$$\Rightarrow 5a = 7.98 \Rightarrow \boxed{a = 1.596}$$

$$\therefore y = 1.596 + 0.432x + 1.428x^2$$

■ Regression

When 2 variables are correlated we can express one variable in terms of the other that means there is an equation connecting the two variables. The line of best fit is also called regression line.

Regression equation X on Y

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

and $\bar{x} = \frac{\sum x}{n}$, $\bar{y} = \frac{\sum y}{n}$

From this

From the following data, obtain Regression equations.

x : 6 2 10 4 8

y : 9 11 5 8 7

x	y	x^2	y^2	xy
6	9	36	81	54
2	11	4	121	22
10	5	100	25	50
4	8	16	64	32
8	7	64	49	56
30	40	220	340	214

Regression equation Y on X

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$r = \sqrt{b_{xy} \cdot b_{yx}}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{30}{5} = 6$$

$$\bar{y} = \frac{40}{5} = 8$$

Regression equation x on y

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

$$\Rightarrow x - 6 = -1.3 (y - 8)$$

$$\Rightarrow x - 6 = -1.3y + 10.4$$

$$\boxed{x = -1.3y + 16.4}$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

$$= \frac{5 \times 214 - 30 \times 40}{5 \times 340 - 40^2}$$

$$= \frac{-130}{100} = -1.3 //$$

Regression equation y on x

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$(y - 8) = -0.65 (x - 6)$$

$$y - 8 = -0.65x + 3.9$$

$$y = -0.65x + 3.9 + 8$$

$$\boxed{y = -0.65x + 11.9}$$

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{5 \times 214 - 30 \times 40}{5 \times 220 - 30^2}$$

$$= \frac{-130}{200} = -0.65 //$$